

MR5029914 05C15 05C10

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Degree-truncated choice number of planar graphs. (English. English summary)

Discrete Math. **349** (2026), no. 7, Paper No. 115035, 15 pp.

All graphs considered are simple and undirected. For a positive integer k , a graph G is said to be *degree-truncated k -choosable* if G is f -choosable for $f(v) = \min\{k, d_G(v)\}$ for each $v \in V(G)$. Note that a connected graph is not degree-choosable if and only if it is a Gallai tree, so such graphs are also not degree-truncated k -choosable for any positive integer k . A connected graph that is not a Gallai tree is degree-truncated Δ -choosable.

Let $\mathcal{P}$ be the class of 3-connected non-complete planar graphs. Richter asked [cf. J. P. Hutchinson, *J. Graph Theory* **59** (2008), no. 1, 59–74; MR2432887] whether every graph in $\mathcal{P}$ is degree-truncated 6-choosable. This was answered in the negative by H. Zhou, J. Zhu and X. Zhu [*J. Combin. Theory Ser. B* **175** (2025), 171–186; MR4933734], who constructed a graph in $\mathcal{P}$ that is not degree-truncated 7-choosable. They also showed that every graph in $\mathcal{P}$ is degree-truncated 16-choosable (in fact, degree-truncated 16-DP-colorable).

In this paper, the authors prove that there exists a graph in $\mathcal{P}$ that is not degree-truncated 8-choosable, and that every graph in $\mathcal{P}$ is degree-truncated 12-choosable. The lower bound is proved by an explicit construction of a 3-connected non-complete planar graph which is not degree-truncated 8-choosable. The authors remark that the proof of the upper bound specifically takes advantage of the list-assignment, and is thus not applicable to DP-colorings, on-line list colorings, or Alon-Tarsi orientations. The authors also announce that the range of k for degree-truncated k -choosability for graphs in $\mathcal{P}$ has been narrowed from $[9, 12]$ to $[10, 11]$ in a subsequent work by H. Xu, Zhou, Zhu and Zhu.

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Note: This list reflects references listed in the original paper as accurately as possible with no attempt to correct errors.